SQL INTERVIEW PREP

20 TRICKY SQL JOIN PATTERNS

Common Traps & Advanced Logic

1. Fill Missing Dates
2. Team Pairings
3. Bought A AND B
4. Bought A NOT B
5. Same Birth Month
6. Update via Join
7. Full Join Union
8. Symmetric Diff
9. Running Total
10. "Top 1" Join
11. Null Key Join
12. LIKE Join
13. Delete Duplicates
14. Many-to-Many
15. Recursive CTE
16. Lateral / Apply
17. Conditional Join
18. Having vs Where
19. Join vs Exists
20. Fan-Out Trap

Q1

FILL MISSING DATES

Report 0 for days with no sales

Standard SQL

```sql
-- Generate all Date x Product combos
SELECT d.date, p.product,
  COALESCE(s.sales, 0)
FROM calendar d
CROSS JOIN products p
LEFT JOIN sales s
  ON d.date = s.date AND p.id = s.pid;
```

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Q2

TEAM PAIRINGS

Unique matches (A vs B, not B vs A)

Standard SQL

```sql
-- Use inequality to avoid mirrors
SELECT a.team_name, b.team_name
FROM teams a
JOIN teams b
  ON a.id < b.id;
```

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Q3

BOUGHT A AND B

Customers who bought both items

Standard SQL

```sql
-- Self Join on Customer ID
SELECT t1.cust_id
FROM orders t1
JOIN orders t2
  ON t1.cust_id = t2.cust_id
WHERE t1.item = 'Milk'
  AND t2.item = 'Bread';
```

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Q4

BOUGHT A NOT B

Bought Phone but NO Cover

Standard SQL

```sql
-- Left Join + Is Null
SELECT t1.cust_id
FROM orders t1
LEFT JOIN orders t2
  ON t1.cust_id = t2.cust_id
  AND t2.item = 'Cover'
WHERE t1.item = 'Phone'
  AND t2.cust_id IS NULL;
```

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Q5

SAME BIRTH MONTH

Find emps who share birthdays

Standard SQL

```sql
-- Join on Month part of date
SELECT a.name, b.name
FROM employees a
JOIN employees b
  ON MONTH(a.dob) = MONTH(b.dob)
  AND a.id <> b.id;
```

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Q6

UPDATE VIA JOIN

Deduct stock based on Sales

SQL Server / MySQL

```sql
-- Update Target joined with Source
UPDATE i
SET i.qty = i.qty - s.sold_qty
FROM inventory i
JOIN daily_sales s
  ON i.product_id = s.product_id;
```

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Q7

FULL JOIN SIM

For DBs without FULL OUTER (MySQL)

MySQL

```
-- Left Join UNION Right Join
SELECT * FROM a LEFT JOIN b ON a.id=b.id
UNION
SELECT * FROM a RIGHT JOIN b ON a.id=b.id;
```

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Q8

SYMMETRIC DIFF

Items in A or B, but NOT both

Standard SQL

```sql
-- Full Join where one side is NULL
SELECT *
FROM table_a a
FULL OUTER JOIN table_b b
  ON a.id = b.id
WHERE a.id IS NULL
   OR b.id IS NULL;
```

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Q9

RUNNING TOTAL

The Self-Join Method

Standard SQL

```sql
-- Inequality Join (Expensive!)
SELECT a.date, SUM(b.amt)
FROM sales a
JOIN sales b
  ON b.date ≤ a.date
GROUP BY a.date;
```

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Q10

"TOP 1" JOIN

Finding Max without Aggregates

Standard SQL

```sql
-- Row where no greater value exists
SELECT a.*
FROM items a
LEFT JOIN items b
  ON b.price > a.price
WHERE b.price IS NULL;
```

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Q11

NULL KEY JOIN

Joining when IDs can be NULL

Standard SQL

```sql
-- Standard '=' fails on NULLs
SELECT * FROM table_a a
JOIN table_b b
  ON COALESCE(a.id, 0) = COALESCE(b.id, 0);
```

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Q12

LIKE JOIN

Matching partial strings

Standard SQL

```
-- Join log to domain rule
SELECT u.email, d.category
FROM users u
JOIN domains d
  ON u.email LIKE CONCAT('%', d.domain);
```

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Q13

DELETE DUPLICATES

Keeping only the first occurrence

MySQL / SQL Server

```sql
-- Self Join Delete
DELETE t1
FROM my_table t1
JOIN my_table t2
  ON t1.email = t2.email
WHERE t1.id > t2.id;
```

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Q14

MANY-TO-MANY

Using a Bridge/Junction Table

Standard SQL

```sql
-- Student -> Enrollment -> Class
SELECT s.name, c.class_name
FROM students s
JOIN enrollment e ON s.id = e.sid
JOIN classes c ON e.cid = c.id;
```

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Q15

RECURSIVE CTE

Finding all subordinates

Standard SQL

```sql
-- Join CTE back to Employee table
WITH RECURSIVE hier AS (
  SELECT id FROM emp WHERE id = 1
  UNION ALL
  SELECT e.id FROM emp e
  JOIN hier h ON e.mgr_id = h.id
) SELECT * FROM hier;
```

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Q16

LATERAL JOIN

Join table to subquery PER ROW

PostgreSQL

```
-- Top 3 comments per user
SELECT u.name, c.text
FROM users u
CROSS JOIN LATERAL (
  SELECT text FROM comments
  WHERE user_id = u.id
  LIMIT 3
) c;
```

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Q17

CONDITIONAL JOIN

Join A or B based on Type

Standard SQL

```sql
-- Polymorphic Join Logic
SELECT e.event_name,
  COALESCE(u.name, c.name)
FROM events e
LEFT JOIN users u
  ON e.rel_id = u.id AND e.type = 'User'
LEFT JOIN companies c
  ON e.rel_id = c.id AND e.type = 'Co';
```

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Q18

HAVING vs WHERE

Filtering Aggregates

Standard SQL

```sql
-- Filter AFTER grouping
SELECT c.name, COUNT(o.id)
FROM customers c
JOIN orders o ON c.id = o.cid
GROUP BY c.name
HAVING COUNT(o.id) > 5;
```

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Q19

JOIN vs EXISTS

Checking presence efficiently

Standard SQL

```sql
-- Exists stops at first match
SELECT * FROM orders o
WHERE EXISTS (
  SELECT 1 FROM products p
  WHERE p.id = o.pid
  AND p.category = 'Tech'
);
```

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Q20

FAN-OUT TRAP

Accidentally exploding rows

Standard SQL

```sql
-- 1:Many joins duplicate left rows
SELECT c.name, COUNT(*)
FROM customers c
JOIN orders o ON c.id = o.cid
-- Result count is Orders, NOT Customers!
GROUP BY c.name;
```

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You Made It!

Learned a New Pattern?

Help your friends learn this too!

REPOST THIS CHEATSHEET

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